Curriculum Vitae

Updated 2026-11-06

GENERAL INFORMATION

Benjamin T. Schelske

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EDUCATION

Ph.D. Analytical Chemistry

Iowa State University
Advised by Robbyn K. Anand
In progress

B.A. in ACS Chemistry, minor in Mathematics

Concordia College Moorhead
2021

SKILLS

Single-Cell & Bioanalytical Methods: Dielectrophoresis (DEP), Label-free cell isolation, Rare and circulating tumor cell analysis, Single-cell enzyme and protease activity assays, Fluorogenic assays, Mammalian cell culture (BSL-2), Experimental design

Analytical Chemistry: Electrochemistry, Electrochemical impedance spectroscopy (EIS), Spectroscopy, Microscopy and quantitative imaging, SPME-GC/MS

Microfabrication & Microfluidics: Microfluidic device design, Cleanroom photolithography, Wafer development, Wet etching, Soft lithography (PDMS), Plasma cleaning

Machine Learning & Computer Vision: PyTorch, TensorFlow/Keras, Scikit-learn, OpenCV, Convolutional neural networks, Transfer learning, Random forests, SVM, PCA, Regression, Cross-validation, Hyperparameter tuning, Imbalanced data (augmentation, SMOTE), Model interpretation (SHAP)

Lab Automation & Image Analysis: Automated image-processing pipelines, Particle tracking, Instrument control and data acquisition (Python serial control, LabVIEW), ImageJ/Fiji macros

Programming & Data: Python (pandas, NumPy, SciPy), MATLAB, SQL, Spark, Statistical analysis, Excel, Rust, HTML/CSS, React

Modeling & Design: COMSOL Multiphysics, AutoCAD, Blender, 3D printing

Communication & Teaching: Scientific writing and publishing, Conference presentations, University teaching, Undergraduate research mentorship, Conference organization

SELECTED PROJECTS

Deep Learning for Tumor Cell Classification (PyTorch, CNN, Transfer Learning): Built and validated CNN pipeline for 1k+ single-cell images from 10+ blood donors, achieving 99.6% accuracy with 5- fold cross-validation. Performed data augmentation to generate synthetic data of minority classes and used SHAP for model interpretation.

Development of Microfluidic Devices for Individual Cancer Cell Analysis (AutoCAD, Microfabrication, Microscopy, ImageJ, Pandas): Designed and fabricated a microfluidic device for label-free cancer cell analysis, designed experiments and established a protocol for imaging and analysis. Performed simulations in COMSOL to aid device design and guide experimental parameters.

Automated Image Processing Pipeline (Python, OpenCV): Developed an automated image processing pipeline for cancer cell analysis using computer vision and particle tracking, reducing manual data analysis time by 8x.

Automated Data Collection and Control of Microfluidic Devices (Python): Created an automated workflow for data collection and manipulation of a microfluidic platform. A user interface and serial control of the microfluidic device improves experimental reproducibility and user productivity.

Random Forest Machine Learning Approach for Microfluidic Quality Control (Python, EIS): Trained a random forest model on electrochemical impedance spectroscopy (EIS) data of microfluidic devices to assess device quality for subsequent experiments. The optimized random forest model increases positive experimental outcomes and improves resource efficiency.

PUBLICATIONS

4. *Marker-Free Classification of Melanoma Cells by Machine Learning Following Their Isolation From Peripheral Blood by Dielectrophoresis*
Benjamin Schelske, Brittany Ginnaty, Han Chen, Josué Rodriguez, Robbyn Anand
ChemRxiv. 2025. 10.26434/chemrxiv-2025-z6k68

3. *Evaluation of Single-Cell Heterogeneity and Invasive Potential in Cancer Cells via Secreted Protease Activity Assay*
Benjamin T Schelske, Ethan Leung, Joseph Banovetz, Jared Anderson, Robbyn K Anand
Analytica Chimica Acta. 2025. 10.1016/j.aca.2025.344663

2. *Label-Free Selection of Cancer Cells by Dielectrophoresis and Evaluation of Invasive Potential by Single-Cell Protease Secretion Assay*
Benjamin T Schelske, Ethan Leung, Joseph Banovetz, Jared Anderson, Robbyn K Anand
ChemRxiv. 2024. 10.26434/chemrxiv-2024-j88pm

1. *Parallel Dielectrophoretic Capture, Isolation, and Electrical Lysis of Individual Breast Cancer Cells to Assess Variability in Enzymatic Activity*
Joseph T Banovetz, Sivani Manimaran, Benjamin T Schelske, Robbyn K Anand
Analytical Chemistry. 2023. 10.1021/acs.analchem.3c00078

TEACHING AND MENTORSHIP

Iowa State University Chemistry TA

Semester	Course	Role / Description	Students
Fall 2026	Graduate Electrochemistry	Head TA + Lecture Substitution	21
Fall 2026	General Chemistry I	Recitation + Lab TA	30
Spring 2026	Physical Chemistry	Teaching Assistant	41
Spring 2025	College Chemistry	Head TA	278
Spring 2023	College Chemistry	Recitation + Lab TA	52
Fall 2022	General Chemistry I	Head TA + Lecture Substitution	362
Spring 2022	General Chemistry II	Online Head TA	112

Semester	Course	Role / Description	Students
Fall 2021	General Chemistry I	Recitation + Lab TA	94
			Total: 990

Mentorship Experience

Semester	Experience	Mentees
Spring 2025	Volunteer Judge - State Science and Technology Fair of Iowa	NA
Spring 2025	CHEM 1630 Head TA	8
Fall 2022 - Fall 2026	Anand Lab Undergraduate-Research Mentor	6
Fall 2022	Graduate Student Orientation TA	46
Fall 2022	CHEM 177 Head TA	8
Fall 2022	Chemistry Learning Community Study Session Volunteer	10
Spring 2021	NCAA DIII Track and Field Men's Team Captain	40
Spring 2021	Concordia College Chemistry Tutor	4
Fall 2020	Concordia College Chemistry Tutor	4
		Total: 126

PRESENTATIONS

17. *Machine-learning classification of marker-free melanoma cells following isolation from peripheral blood by dielectrophoresis*
Benjamin T. Schelske, Brittany Ginnaty, Han Chen, Josué Rodriguez, Robbyn Anand
June, 2026 - DEP2026, Oral Presentation, Louisville, KY

16. *Functional Phenotyping of Single Cancer Cells Using Dielectrophoretic Isolation And Compartmentalization*
Benjamin T. Schelske, Joseph T. Banovetz, Darshna Pagariya, Ethan H. Leung, Sivani Manimaran, Quinlan G. Pollak, Jared L. Anderson, Robbyn K. Anand
June, 2026 - DEP2026, Work Presented, Louisville, KY

15. *Monitoring Key Performance Signatures in a Microscale Single-Cell Analysis Device in Real Time by Machine Learning*
Benjamin T. Schelske, Anuj Sharma, Robbyn K. Anand*
April 16, 2025 - TrAC Day 2025, Video Presentation, Iowa State University, Ames, IA

14. *Dielectrophoretic Isolation of Melanoma Cells Followed by Single-Cell Protease Activity to Evaluate Invasive Potential*
Benjamin T. Schelske, Ethan H. Leung, Joseph T. Banovetz, Jared L. Anderson, Robbyn K. Anand*
October 16, 2024 - µTAS 2024, Poster Presented, Montréal, QC, Canada

13. *Label-Free Selection of Cancer Cells by Dielectrophoresis and Evaluation of Invasive Potential by Single-Cell Protease Secretion Assay*
Benjamin T. Schelske, Ethan H. Leung, Joseph T. Banovetz, Jared L. Anderson, Robbyn K. Anand*
August 19, 2024 - ACS Fall 2024 Rising Stars in Analytical Chemistry, Oral Presentation, Denver, CO

12. *Characterizing Dielectrophoretic (DEP) Response of Melanoma Cell Lines With Known Drug Resistance*
Pallavi Baliyara, Brittany Ginnaty, Olivia Jozaitis, Benjamin Schelske, Thilini Rathnaweera, Christopher Stipp, Michael Henry, Robbyn K. Anand*
August 2, 2024 - Summer Undergraduate Research Symposium, Poster Presented, Iowa State University, Ames, IA

11. *Identification of A375 Cell Line Drug Resistance Using Dielectrophoresis*
Olivia Jozaitis, Brittany Ginnaty, Pallavi Baliyara, Thillini Rathnaweera, Benjamin Schelske, Michael Henry, Christopher Stipp, Robbyn K. Anand*
August 2, 2024 - Summer Undergraduate Research Symposium, Poster Presented, Iowa State University, Ames, IA

10. *Comparative Performance of Density Gradient Centrifugation and Immunodensity Cell Isolation Sample Debulking Techniques for Subsequent On Chip Single Cell Analysis*
Brittany C. Ginnaty, Benjamin T. Schelske, Robbyn K. Anand*
June 28-30, 2024 - Midwest Retreat for Diversity in Chemistry (MWRDC), Poster Presented, Bradley University, Peoria, IL

9. *Label-Free Selection of Cancer Cells by Dielectrophoresis and Evaluation of Invasive Potential by Single-Cell Protease Secretion Assay*
Benjamin T. Schelske, Ethan H. Leung, Joseph T. Banovetz, Jared L. Anderson, Robbyn K. Anand*
April 24, 2024 - Cotton-Uphaus Competition, Oral Presentation, Iowa State University, Ames, IA

8. *Evaluation of Dielectrophoretic Response of Naïve and Drug-Resistant Melanoma*
Josué Rodríguez, Benjamin T. Schelske, Robbyn K. Anand*
April 8-10, 2024 - National Conference on Undergraduate Research (NCUR) 2024, Poster Presented, Long Beach, CA

7. *Label-Free Selection of Cancer Cells by Dielectrophoresis and Evaluation of Invasive Potential by Single-Cell Matrix Metalloprotease-9 Activity Assay*
Benjamin T. Schelske, Ethan H. Leung, Joseph T. Banovetz, Jared L. Anderson, Robbyn K. Anand*
November 3, 2023 - Midwestern Universities Analytical Chemistry Conference (MUACC), Poster Presentation, Iowa City, IA

6. *Sample Debulking via Density Gradient Centrifugation and Immunodensity Cell Isolation for Improving On Chip Single Cell Analysis*
Brittany C. Ginnaty, Benjamin T. Schelske, Robbyn K. Anand*
June 23-25, 2023 - Midwest Retreat for Diversity in Chemistry (MWRDC), Poster Presented, Parkville, MO

5. *Measuring the Invasive Potential of Cancer Cells with a Small Inexpensive Device*
Benjamin T. Schelske, Ethan H. Leung, Joseph T. Banovetz, Jared L. Anderson, Robbyn K. Anand*
October 27, 2022 - 3MT: Three Minute Thesis Preliminary Heat, Iowa State University, Ames, IA

4. *Label-Free Selection of Circulating Tumor Cells by Dielectrophoresis and Evaluation of Invasive Potential by Single-Cell Protease Activity Assay*
Benjamin T. Schelske, Ethan H. Leung, Joseph T. Banovetz, Jared L. Anderson, Robbyn K. Anand*
August 21-25, 2022 - ACS Fall Technical Program, Poster Presentation, Chicago, IL

3. *Data Preparation and Analysis Automation of a Fluorogenic Assay Using ImageJ Macros and MATLAB Scripting*
Benjamin T. Schelske, Ethan H. Leung, Joseph T. Banovetz, Jared L. Anderson, Robbyn K. Anand*
June 24-26, 2022 - Midwest Retreat for Diversity in Chemistry (MWRDC), Poster Presentation, Grinnell, IA

2. *Developments in the Analysis of Exhaled Breath as a Means of COVID-19 Detection*
Benjamin T. Schelske, Mark B. Jensen*
February 26, 2021 - Senior Seminar, Oral Presentation, Concordia College, Moorhead, MN

1. *Determination of Phthalate Esters in Minnesota Water Bodies Using Head Space Solid-Phase Microextraction*
Benjamin T. Schelske, Jenna L. Pauly, Mark B. Jensen*
April 10-13, 2021 - Celebration of Student Success (COSS), Poster Presentation, Concordia College, Moorhead, MN

AWARDS

Dielectrophoresis Society Travel Award
June, 2026 - Dielectrophoresis Society

The Frank J. Moore and Thoreen Beth Moore Fellowship
April 15, 2026 - Iowa State University, Ames, IA

Mary K. and Velmer A. Fassel Analytical Chemistry Fellowship
April 23, 2025 - Iowa State University, Ames, IA

Cotton-Uphaus Oral Presentation Finalist
April 24, 2024 - Iowa State University, Ames, IA

Iowa State University Teaching Excellence Award
May 11, 2023 - Iowa State University, Ames, IA

PROFESSIONAL SERVICE

Dielectrophoresis 2026 Co-Chair
June 24-26, 2026 - University of Louisville, Louisville, KY

CERTIFICATION

The Last Algorithms Course You'll Need
December 2, 2025

Pittsburgh Supercomputing Center: HPC Monthly Workshop: Machine Learning and BIG DATA
October 14-15, 2025

LinkedIn Learning: SQL Essential Training
December 31, 2024

ACS Meetings & Expositions Presenter
September 17, 2024

American Chemical Society Member
July 11, 2024

NVIDIA DLI Certificate, Fundamentals of Accelerated Computing with CUDA C/C++
October 9, 2023